

1 Population estimand

$$\theta = I(C; B \mid \mathcal{F}^0)$$

$$H(C \mid \mathcal{F}^0) - H(C \mid \mathcal{F}^B) = \theta$$

Orders population DELB/Fano floors

Predictive information; not a causal effect



sample + discretize

2 Finite-sample estimator

$$\hat{\theta} = \theta + b(N, K_{\text{obs}}, S_1, \nu) + \varepsilon$$

Plugin or Miller--Madow statistic

Bias and variance depend on support

K_{obs}/N , S_1 , ν/N diagnose sparsity



apply π ; recompute

3 Randomization reference

$$\mathcal{L}_{\pi}(\hat{\theta}^{\pi} \mid H_0)$$

$\mu_{\text{null}} = \mathbb{E}_{\pi}[\hat{\theta}^{\pi}]$ need not be zero

Compare $\hat{\theta}_{\text{obs}}$ with its null law

Recompute statistic and support after π